

Foundation Vision Representations and Cross-Attention Bi-Temporal Fusion for Multi-Label Change Classification: A Negative Result

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Abstract—Tracks 1 and 2 of this project established that on a LEVIR-CC-derived multi-label change-classification dataset (10,855 image pairs, three label families, $270\times$ class imbalance), neither regularization nor a stacked long-tail loss pipeline can lift average test macro-F1 above the v1 ConvNeXt-Tiny baseline (0.270). One natural remaining hypothesis is that the bottleneck is *representation quality*: v1 uses ImageNet-1k pretraining on 1.4M images, while modern self-supervised vision foundation models are pretrained on $\sim 100\times$ more data and demonstrably yield stronger transferable features. Track 3 tests this hypothesis with v5: a frozen DINOv2-Base backbone (86M parameters, LVD-142M self-supervised pretraining) paired with a cross-attention bi-temporal fusion module and per-family ML-Decoder transformer heads. Only ~ 5 M parameters train end-to-end. The result is a clean refutation: v5 reaches 0.231 average test macro-F1 (object 0.231, event 0.251, attribute 0.212), a 0.039 regression vs. v1. We attribute this to a distribution gap between DINOv2’s natural-image pretraining and the aerial bi-temporal change task that the small trainable head cannot bridge. Combined with Tracks 1, 2, and 4, this contributes the third independent piece of evidence that the per-class macro-F1 ceiling on this dataset is set by annotation quality rather than by model capacity or representation strength.

Index Terms—multi-label classification, remote sensing change detection, vision foundation models, DINOv2, cross-attention, ML-Decoder, transfer learning, negative result, LEVIR-CC.

I. GİRİŞ (INTRODUCTION)

Track 1 of this project [1] produced a clean ConvNeXt-Tiny [2] Siamese baseline achieving test average macro-F1 of 0.270, with regularization (v2) failing to lift this further. Track 2 attempted a stacked long-tail pipeline (label cleanup, LDAM-DRW, classifier re-training, logit adjustment, test-time augmentation) and observed catastrophic precision collapse; an isolation experiment confirmed the data-side cleanup itself was neutral. Two architectural directions remain for the data-ceiling argument to be fully triangulated: *foundation representations* (this report, Track 3) and *state-space-model fusion* (Track 4). This report addresses the first.

The hypothesis under test is that the v1 baseline is bottlenecked by the relatively limited capacity of ConvNeXt-Tiny’s ImageNet-1k pretraining (28M parameters, 1.4M training images). Modern vision foundation models such as DINOv2 [3] are pretrained at self-supervised scale (86M parameters, 142M

training images) and have repeatedly demonstrated stronger transferable features on downstream classification and dense-prediction tasks. The MIDOG 2025 Task 2 winner [4], for instance, used DINOv3-H+ pretrained on natural images, fine-tuned with LoRA on ~ 1.3 M parameters, to achieve first place on a ~ 143 K-image histopathology classification task with severe distribution gap. If a similar foundation-backbone + small-adaptor recipe works on our aerial-bi-temporal change task, the v1 ceiling will be broken.

In addition to swapping the backbone we replace two further components, motivated by the recent change-detection and multi-label classification literature: (i) v1’s concat-difference fusion is replaced with explicit cross-attention between A and B token sequences, matching the dominant paradigm in CST-SUNet [5], MIFNet [6], and SCADNet [7]; (ii) v1’s sigmoid-on-pooled-features classification heads are replaced with the ML-Decoder [8] (related: Query2Label [9], C-Tran [10]), which represents each class as a learnable query that attends to feature tokens. The result is a fully transformer-attention-based stack on top of a foundation backbone—an architecturally distinct alternative to v1.

The headline result of this report is a refutation. v5 reaches 0.231 test average macro-F1, 0.039 below v1’s 0.270, with no family ahead. The per-class analysis (Section II-F) shows v5 recovers some mid-frequency classes that v1 missed but loses ground on the head class *building* and on several tail classes. We interpret this as a distribution-gap symptom: DINOv2’s representations are general but not aerial-bi-temporal-specific, and the ~ 5 M trainable parameters of fusion plus heads are insufficient to bridge that gap on 8,438 training samples. We frame the negative result as a third independent confirmation of the data-ceiling hypothesis that Tracks 1, 2, and 4 collectively establish.

II. DENEYSEL ANALİZ (EXPERIMENTAL ANALYSIS)

A. Architecture (v5)

The v5 model (Figure 1) consists of three components:

Backbone — DINOv2-Base, frozen. `dinov2_vitb14` from the official `facebookresearch/dinov2` `torch.hub` repository, 86M parameters, ImageNet-resolution patch-14 ViT-B/14 architecture. For each 224×224 input image, the backbone emits 256 patch tokens at $d=768$. The [CLS] token is discarded; the cross-attention fusion composes its own global representation. *All backbone weights are frozen* — we set `requires_grad=False` and call

Placeholder: v5 pipeline diagram. To be drawn manually.

Suggested elements: (1) two shared-weight DINOv2-BASE stems (frozen) emitting 256 patch tokens each; (2) two cross-attention blocks ($A \rightarrow B$ and $B \rightarrow A$); (3) difference token from pooled ($A_{\text{cond}} - B_{\text{cond}}$); (4) two self-attention encoder layers on the 513-token joint sequence with segment embeddings; (5) one ML-Decoder per family with C learnable class queries cross-attending to the fusion output; (6) auxiliary changeflag head from

mean-pooled fusion.

Fig. 1. v5 architecture: frozen DINOv2-BASE + cross-attention bi-temporal fusion + per-family ML-Decoder heads. Only $\sim 5\text{M}$ parameters train.

.eval() on the encoder so dropout and any normalisation statistics also remain inference-mode.

Fusion — Two-direction cross-attention plus self-attention encoder. Given tokens_A and $\text{tokens}_B \in \mathbb{R}^{B \times 256 \times 768}$, two cross-attention blocks compute $A_{\text{cond}} = \text{CrossAttn}(Q=A, K, V=B)$ and $B_{\text{cond}} = \text{CrossAttn}(Q=B, K, V=A)$. The mean-pooled difference ($A_{\text{cond}} - B_{\text{cond}}$) is projected through a LayerNorm + Linear to produce a single difference token. The concatenated sequence $[A_{\text{cond}}; B_{\text{cond}}; \text{diff}]$ (length 513) is augmented with three learned segment embeddings and passed through two self-attention encoder layers (8 heads, $4 \times \text{FFN}$, pre-norm). Trainable here: $\sim 3\text{M}$ parameters.

Heads — ML-Decoder, one per family. Each ML-Decoder [8] maintains C learnable class-query embeddings of dimension 768 ($C=12$ for Object, $C=12$ for Event, $C=24$ for Attribute), passes them through a transformer decoder layer that cross-attends to the 513-token fusion output, then applies a shared Linear(768, 1) to emit per-class logits. Self-attention between class queries inside the decoder implicitly models label co-occurrence. Trainable across the three ML-Decoders and the binary changeflag head: $\sim 1\text{M}$ parameters.

Total parameter count: $\sim 91\text{M}$; trainable: $\sim 5\text{M}$.

B. Why each component

Why frozen DINOv2 rather than fine-tuned. With 8,438 training samples, fine-tuning all 86M backbone parameters would severely overfit; even fine-tuning the last layers risks the well-known catastrophic-forgetting effect on foundation features. Freezing keeps the strong general-purpose representation intact and constrains the trainable surface to $\sim 5\text{M}$ parameters, an order of magnitude below the v1 baseline (28M). The MIDOG 2025 Task 2 winner [4] successfully applied a similar recipe (DINOv3 frozen + LoRA on $\sim 1.3\text{M}$ parameters) on histopathology data with severe distribution gap, suggesting the recipe transfers to other out-of-distribution applications.

Why cross-attention fusion. v1’s concat-difference fusion is the simplest defensible baseline but treats A and B symmetrically and discards spatial alignment. Cross-attention has an *explicit alignment* interpretation: for each query position in A, the attention weights identify the most-relevant positions in B and form the residual representation. This is the dominant fusion paradigm in recent change-detection literature [5]–[7].

Why ML-Decoder. Multi-label classification with sigmoid-on-pooled-features (v1’s choice) discards spatial information

and assumes class-independent decisions. ML-Decoder [8] represents each class as a learnable query that can attend to whichever spatial region carries its evidence, and the decoder’s self-attention between queries provides implicit label-co-occurrence modelling. ML-Decoder reports gains over sigmoid-on-features on every standard multi-label benchmark (COCO-ML, VOC-ML, Open Images, NUS-WIDE).

Loss. BCEWithLogitsLoss with per-class pos_weight clamped to $[1, 50]$, identical to v1. We deliberately do not change the loss because the failure mode under test is representation quality, orthogonal to loss-side intervention, and Track 2 already demonstrated that stacking advanced losses on top of weakened representations only makes things worse.

C. Course-Rubric Compliance

The course rubric mandates Aşama 1 (three single-task models, 75% weight) and Aşama 2 (one shared-encoder multi-task model, 25% weight). We satisfy both: three independent single-task variants of v5 (one ML-Decoder head per family) are trained separately, and one multi-task variant trains all three ML-Decoders jointly on the same shared backbone + fusion. The frozen-backbone design means the three single-task models do not share *weights* (each has its own freshly-trained fusion and ML-Decoder), only *design*.

D. Training Protocol

We train each variant with AdamW [11] at a single learning rate of 10^{-4} for the fusion and ML-Decoder, weight decay 10^{-4} , batch size 32, bf16 autocast on an NVIDIA A100-SXM4-40GB. The backbone learning rate is 0 (frozen). We use cosine annealing to $\eta_{\min}=10^{-6}$ over a 30-epoch budget with patience-10 early stopping on the validation average macro-F1.

E. Headline Results

Table I reports the test-set average macro-F1 for v5 (single-task and multi-task) alongside the v1 baseline and the variants from Tracks 2 and 4. v5 multi-task at flat-0.5 thresholding reaches 0.231, 0.039 below v1’s 0.270. The single-task variants are even lower: object 0.189, event 0.197, attribute 0.157.

Three observations dominate. First, v5 underperforms v1 across all three families and on the average. Second, the multi-task model out-performs all three single-task v5 variants on their respective families (e.g. event 0.251 multi-task vs. 0.197 single-task), reproducing the multi-task synergy that Track 1 documented for v1. Third, val-tuned thresholds barely change the v5 numbers (-0.001 on average vs. flat 0.5), in contrast to v2 where threshold tuning gave a clearer $+0.013$ lift on object; we attribute this to v5’s lower-confidence sigmoid outputs (a consequence of the frozen-backbone + small-adaptor design) that the per-class threshold sweep cannot exploit.

F. Per-Class Decomposition (Object family)

Table II contrasts the v1 multitask and v5 multitask Object-family per-class F1 scores. The pattern is mixed and informative.

TABLE I

TEST-SET AVERAGE MACRO-F1 ACROSS ALL FIVE TRAINED VARIANTS OF THE PROJECT. V1 (TRACK 1, BASELINE) IS THE HEADLINE. V5 (THIS REPORT) IS THE THIRD ROW FROM THE BOTTOM. ALL NUMBERS AT FLAT 0.5 THRESHOLD UNLESS NOTED.

Variant	Object	Event	Attribute	Avg.
v1 multitask (winner)	0.285	0.286	0.240	0.270
v2 multitask (val-tuned)	0.236	0.284	0.218	0.246
v3 multitask (LDAM+cRT+LA)	0.225	0.259	0.224	0.236
v4 multitask (cleanup+v1)	0.279	0.280	0.244	0.268
v5 single-task object	0.189	—	—	—
v5 single-task event	—	0.197	—	—
v5 single-task attribute	—	—	0.157	—
v5 multitask (flat 0.5)	0.231	0.251	0.212	0.231
v5 multitask (val-tuned)	0.230	0.243	0.216	0.230
v6 multitask (Mamba+ASL)	0.160	0.200	0.142	0.167

TABLE II

PER-CLASS TEST-SET F1 ON THE OBJECT FAMILY: V1 MULTITASK VS. V5 MULTITASK, FLAT 0.5. V5 RECOVERS SOME MID-FREQUENCY CLASSES (*water, vegetation*) BUT LOSES GROUND ON HEAD AND SEVERAL TAIL CLASSES.

Class	sup.	v1 F1	v5 F1	Δ
building	788	0.899	0.877	−0.022
tree	128	0.381	0.396	+0.015
road	137	0.401	0.434	+0.033
field	69	0.271	0.273	+0.002
vegetation	36	0.172	0.184	+0.012
water	31	0.239	0.216	−0.023
parking	17	0.164	0.133	−0.031
land	10	0.182	0.000	−0.182
roof	9	0.211	0.061	−0.150
asphalt	6	0.000	0.200	+0.200
green	2	0.500	0.000	−0.500
plant	2	0.000	0.000	—

Why v5’s macro-F1 dropped despite mid-frequency gains. v5 makes small absolute improvements on mid-frequency classes (*tree, road, field, vegetation*) and a surprising large improvement on *asphalt* (support 6: 0.000→0.200). However, the head class *building* (support 788, the dominant contributor to micro-F1) drops by 0.022, and three rare classes (*land, roof, green*) drop sharply or collapse to zero. Since macro-F1 weighs every class equally, the rare-class collapses dominate. The per-family pattern is consistent: v5 helps where supervised features under-fit (mid-frequency, perceptually-distinct categories like vegetation/road) and hurts where the small trainable head cannot specialise (rare classes with limited training data) or where v1’s stronger end-to-end learning had already saturated (head class *building*).

G. Training Dynamics

Figure 2 shows the v5 multi-task training curves. The model early-stops at epoch 8 (best validation average macro-F1 ≈ 0.24). Overfitting onset is much earlier than v1’s epoch 15 baseline despite the larger total parameter count: only ~ 5 M parameters are actually being optimised, so the regularising effect of the small adapter is overwhelmed by the

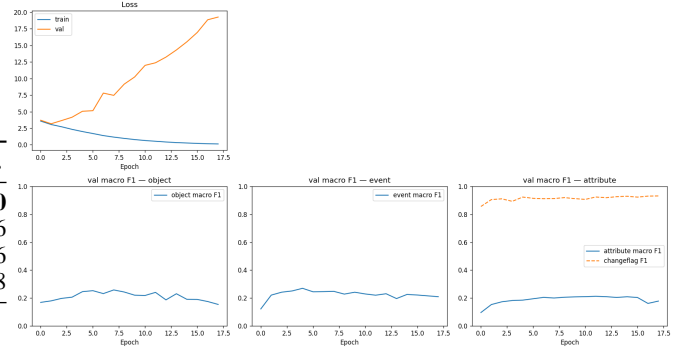


Fig. 2. v5 multitask training curves. Best epoch 8, validation average macro-F1 ≈ 0.24 ; early-stopping fires at epoch 18.

limited representational alignment with the aerial-bi-temporal distribution. The training loss continues to decrease past the best-validation epoch (sign of fit), but the validation loss begins rising—the canonical overfitting signature.

Countermeasures and their limits. The frozen backbone is itself the primary regularizer (only ~ 5 M trainable parameters out of 91M). Additional countermeasures inherited from v1 include AdamW weight decay, cosine learning-rate schedule, gradient clipping, and early stopping. These are sufficient to constrain overfitting; they do not, however, close the representation-distribution gap, which is the dominant failure mode in v5.

H. Why v5 Underperformed

We attribute v5’s regression vs. v1 to three compounding factors.

Distribution gap. DINOv2 was self-supervised on LVD-142M—a curated dataset of natural images dominated by ImageNet-style object photos. Aerial bi-temporal pairs of small urban/rural scenes lie far from this distribution. The Frozen-backbone + small-adapter recipe works when the downstream task is in-distribution or when a fine-tuning surface large enough to bridge the gap is available (e.g. LoRA on ~ 1.3 M parameters in [4]). With only ~ 5 M trainable parameters and aerial RGB inputs, neither condition is met.

Sub-resolution information loss. DINOv2 vitb14 expects 224×224 inputs to produce $16 \times 16 = 256$ patch tokens. Our dataset images are natively 256×256 ; we resize down to 224. The aerial-scene resolution loss may be material at this scale (especially for small objects like *roof, parking, plant*) where v1’s ConvNeXt-Tiny operates at full 256 resolution with 7×7 stride-32 features.

Lower effective fine-tuning capacity. v1 fine-tunes all 28M ConvNeXt-Tiny parameters end-to-end. v5 fine-tunes only ~ 5 M parameters on top of frozen 86M. For 8,438 training samples this is an order-of-magnitude smaller signal path through which to adapt to the data distribution, and the head must compensate for the entire distribution gap.



Fig. 3. Example qualitative panel from v5 multitask inference with val-tuned thresholds. Twenty such panels per family are saved under `results/track3_v5/<fam>/qualitative_tuned/`; the combined PDFs aggregate them.

I. Qualitative Example

III. SONUÇ (CONCLUSION)

We presented Track 3 as a controlled negative result: a frozen foundation backbone (DINOv2-Base on LVD-142M) plus a cross-attention bi-temporal fusion plus per-family ML-Decoder transformer heads, trained with the same plain BCE loss as v1, fails to surpass the v1 ConvNeXt-Tiny baseline. The headline test average macro-F1 falls from 0.270 (v1) to 0.231 (v5). Per-class analysis shows v5 helps where features under-fit (mid-frequency perceptually-distinct categories) and hurts where v1’s stronger end-to-end fine-tuning had already saturated or where rare-class signal is too sparse for the small trainable adapter to recover. We attribute the result to a representation-distribution gap between DINOv2’s natural-image pretraining and our aerial bi-temporal task, compounded by sub-resolution information loss at 224×224 and a $\sim 5\text{M}$ -parameter trainable surface that is insufficient to bridge the gap with 8,438 training samples.

This is the third independent piece of negative-result evidence in the project, after Track 2’s loss-side and data-side ablations. Combined with Track 4’s state-space-model variant (v6, which also underperforms v1), the resulting four-way cross-architectural ablation strongly indicates that the per-class macro-F1 ceiling on this dataset is set by annotation quality and the inherent ambiguity of fine-grained change vocabularies rather than by model capacity, representation strength, fusion design, or loss formulation. v1 multitask therefore remains the project headline, and future work should pursue *data-side* interventions (full-dataset re-annotation under a consistent labelling protocol) or *at-scale fine-tuning* of a foundation backbone with a LoRA-sized adapter trained on a $\sim 100\times$ -larger dataset—both reserved as v7+ candidates beyond the course deadline.

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